Magnetic Torque Investigations

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1 Introduction and Background

Magnetic dipoles are involved in a variety of phenomena in quantum mechanics and particle physics. Their precession in a magnetic field provides the fundamental mechanism for nuclear magnetic resonance (NMR), used in applications from molecular structure analysis to magnetic resonance imaging (MRI) [1,3]. Magnetic dipole precession contextualizes prototypical problems representative of central quantum phenomena, such as the Zeeman effect [1]. This experiment seeks to investigate typical magnetic dipole behaviors associated with torques produced by external magnetic fields. Based on a dipole's simple harmonic motion and precession in a magnetic field, experimental values of the dipole strength are obtained. A qualitative investigation of the spin flip mechanism central to NMR is also conducted.

Consider a dipole placed in a magnetic field with some orientation angle α measured from the dipole vector to the field vector. This dipole experiences a torque $\vec{\tau}$ defined by:

$$\vec{\tau} = \vec{\mu} \times \vec{B} = \mu B \sin\alpha \tag{1}$$

where $\vec{\mu}$ represents the magnetic dipole moment and $\vec{B}$ is the external magnetic field vector [3]. When the dipole is positioned with some angular displacement $\gamma = -\alpha$ measured from $\vec{B}$ to $\vec{\mu}$, the torque produced by the field opposes the initial displacement. This restoring torque is approximately proportional to the displacement for small α, making the motion in the small-α regime simple harmonic [2]. In this regime, Newton's second law for torques gives [3]:

$$\tau = \mu B \sin\alpha = -\mu B \sin\gamma \Rightarrow \frac{d^2\gamma}{dt^2} = -\frac{\mu B}{I}\gamma \tag{2}$$

This is the equation of motion for simple harmonic motion with an angular frequency ω satisfying [2]:

$$\omega^2 = \left(\frac{2\pi}{T}\right)^2 = \mu \cdot \frac{B}{I} \tag{3}$$

where in general the relationship between the angular frequency and the period of simple harmonic oscillatory motion is given by [2]:

$$\omega = \frac{2\pi}{T} \tag{4}$$

Another typical behavior of magnetic dipoles in magnetic fields is precession. Magnetic precession arises when an object has a magnetic dipole $\vec{\mu}$ parallel to the object's angular momentum $\vec{L}$ in a non-parallel magnetic field $\vec{B}$ [1,3]. In this situation, (1) dictates that the torque experienced

by the dipole is orthogonal to the object's angular momentum and to the magnetic field $\vec{B}$. This causes the angular momentum vector to change direction with a constant rate and locks the change in angular momentum $\Delta\vec{L} = \vec{\tau}\Delta t = \frac{d\vec{L}}{dt}\Delta t$ to the plane orthogonal to $\vec{B}$, as in Figure 1.

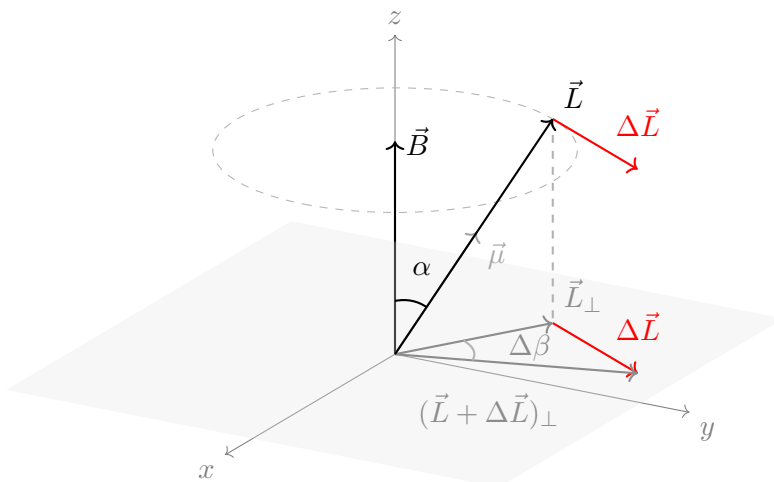

Figure 1: The angular momentum $\vec{L}$ and magnetic dipole moment $\vec{\mu}$ of an object experiencing precession in a magnetic field $\vec{B}$. Note the circular path of the tip of $\vec{L}$, to which the change in angular momentum $\Delta\vec{L}$ caused by the magnetic field torque is tangent. Also note the projection of $\vec{L}$ onto the plane orthogonal to $\vec{B}$, $\vec{L}_\perp$; the projection of $(\vec{L}+\Delta\vec{L})$ onto the same plane, $(\vec{L}+\Delta\vec{L})_\perp$; and the angle $\Delta\beta$ between them. The dipole moment is oriented with an angle α with the magnetic field.

Consider the projection of the object's angular momentum $\vec{L}_\perp$ into the plane orthogonal to the magnetic field, with magnitude given by $L\sin\alpha$. Also consider the projection of the angular momentum after a small change in time Δt: $(\vec{L}+\Delta\vec{L})_\perp$. The angle between these is $\Delta\beta$, a change in the angle of precession β of the object. Considering $\Delta\beta$ to be small and noting the right triangle formed by $\vec{L}_\perp$, $(\vec{L}+\Delta\vec{L})_\perp$, and $\Delta\vec{L}$, a relationship between the precession angular frequency $\omega_p = \frac{d\beta}{dt}$, the spin angular frequency ω_s, and the angle from the magnetic field α is obtained:

$$\Delta L = L\sin\alpha\sin\Delta\beta \Rightarrow \frac{dL}{dt} = L\frac{d\beta}{dt}\sin\alpha \Rightarrow \tau = I\omega_s\omega_p\sin\alpha \tag{5}$$

With equations (1) and (4), this provides a relationship between the period T_s of the object's motion about its angular momentum axis, the angular frequency ω_s of the object's motion about its angular momentum axis, the period T_p of precession, the angular frequency ω_p of precession, the object's moment of inertia I, the dipole strength μ, and the magnetic field strength B:

$$\omega_s\omega_p = \frac{(2\pi)^2}{T_pT_s} = \mu\cdot\frac{B}{I} \tag{6}$$

Note the symmetry with (3).

2 Methodology

A magnetic dipole with known dipole direction but unknown dipole strength μ is constructed. A time-invariant magnetic field $\vec{B}$ that the dipole is placed in is produced by two twin magnetic coils,

as shown in Figure 2. The strength of $\vec{B}$ is controlled by the current through the coils [3] and measured directly with a Gaussmeter at the dipole's position. Dissipative forces due to contact between the dipole and a holder for the dipole are mitigated with a cushion of compressed air, allowing the dipole to rotate freely. The moment of inertia I of the dipole is estimated as that of a solid sphere of mass $m = 183.0 \pm 0.1$ g and radius $r = 28.5 \pm 3$ mm [3].

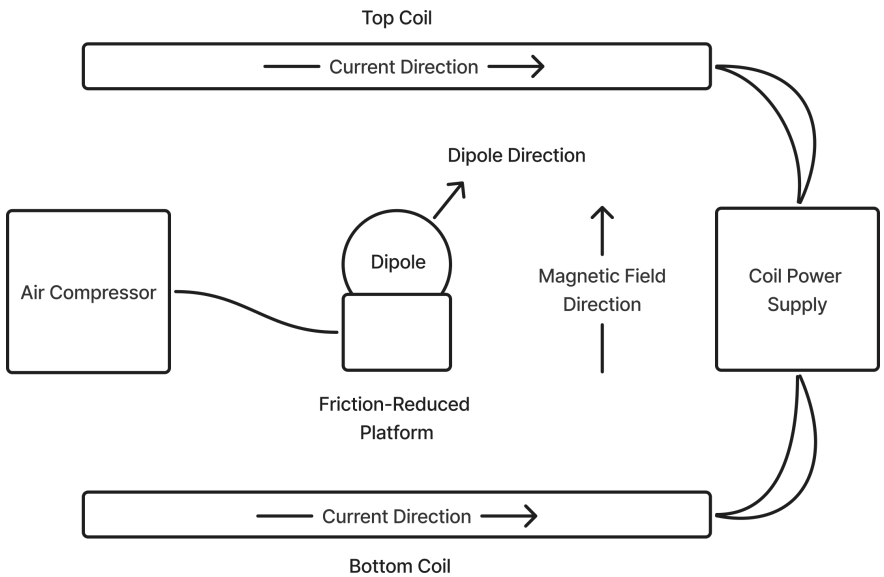

Figure 2: Experimental apparatus for the three magnetic dipole experiments. The dipole platform sits between two magnetic coils, producing an upward magnetic field due their agreeing currents (supplied by a DC power supply). An air compressor provides a low-friction cushion for the dipole.

The unknown strength of the dipole is measured using two experiments: an experiment of simple harmonic motion and an experiment of magnetic precession.

2.1 Simple Harmonic Motion

At each current in Table 1, the dipole is placed on its holder with $\vec{\mu}$ aligned with $\vec{B}$ and given a small initial angular displacement α_0 about $\vec{B}$. It is then allowed to oscillate and the average period across five oscillation cycles is measured with a clock. This five-oscillation average is repeated five times; another average is taken over these repetitions to obtain a final period of oscillation T.

2.2 Magnetic Precession

At each current in Table 1, the dipole is placed upon its holder with $\vec{\mu}$ oriented at an angle α away from the vertical. The dipole is then given a modest angular velocity ω_s (period T_s) about its dipole axis. Precession results, and the precession period T_p is obtained as an average over half a precession cycle. ω_s is derived from an average between readings from a tachometer taken at the start and end of this half-cycle. Note that only half a cycle is considered to minimize energy loss of the dipole due to air resistance.

2.3 Spin Flip Mechanism

A brief exploration of the spin flip mechanism is also conducted. As in the precession experiment, the dipole is placed upon its holder with an initial angular velocity and allowed to precess. Two magnets are then placed on either side of the dipole and rotated so as to appear motionless in the dipole's precessing frame of reference, producing a magnetic field that is constantly perpendicular to the dipole as it precesses. It is observed that the dipole flips its orientation under the torque caused by the external magnetic field, in the direction predicted by equation (1).

3 Results

Data points for magnetic fields corresponding to all currents are presented in Table 1. Equations (3) and (6) suggest that plotting $\left(\frac{2\pi}{T}\right)^2$ and $\frac{(2\pi)^2}{T_p T_s}$ against $\frac{B}{I}$, respectively, should yield linear relationships whose slopes yield two experimental value of the dipole strength μ. These relationships from the experimental data are plotted in Figures 3 and 4 and linear regression is performed, yielding μ-values of (1.63 ± 0.08) A $\cdot$ m^2 and (1.47 ± 0.13) A $\cdot$ m^2, respectively. These values are consistent with each other to within 1σ.

Current (A) ($\pm$ 0.01 A)	T (s)	T_s (s)	T_p (s) ($\pm$ 1 s)	B (10^{-3} T) ($\pm 10^{-5}$ T)
2.00	1.020 ± 0.009	0.0590 ± 0.0012	23.1	1.39
1.75	1.134 ± 0.015	0.062 ± 0.003	24.2	1.17
1.50	1.221 ± 0.016	0.0646 ± 0.0028	30.4	1.01
1.25	1.365 ± 0.022	0.0578 ± 0.0024	43.2	0.83
1.00	1.502 ± 0.008	0.074 ± 0.003	46.9	0.64

Table 1: Measured values of the simple harmonic motion period T, spin period T_s, precession period T_p, and magnetic field strength B for each coil current tested.

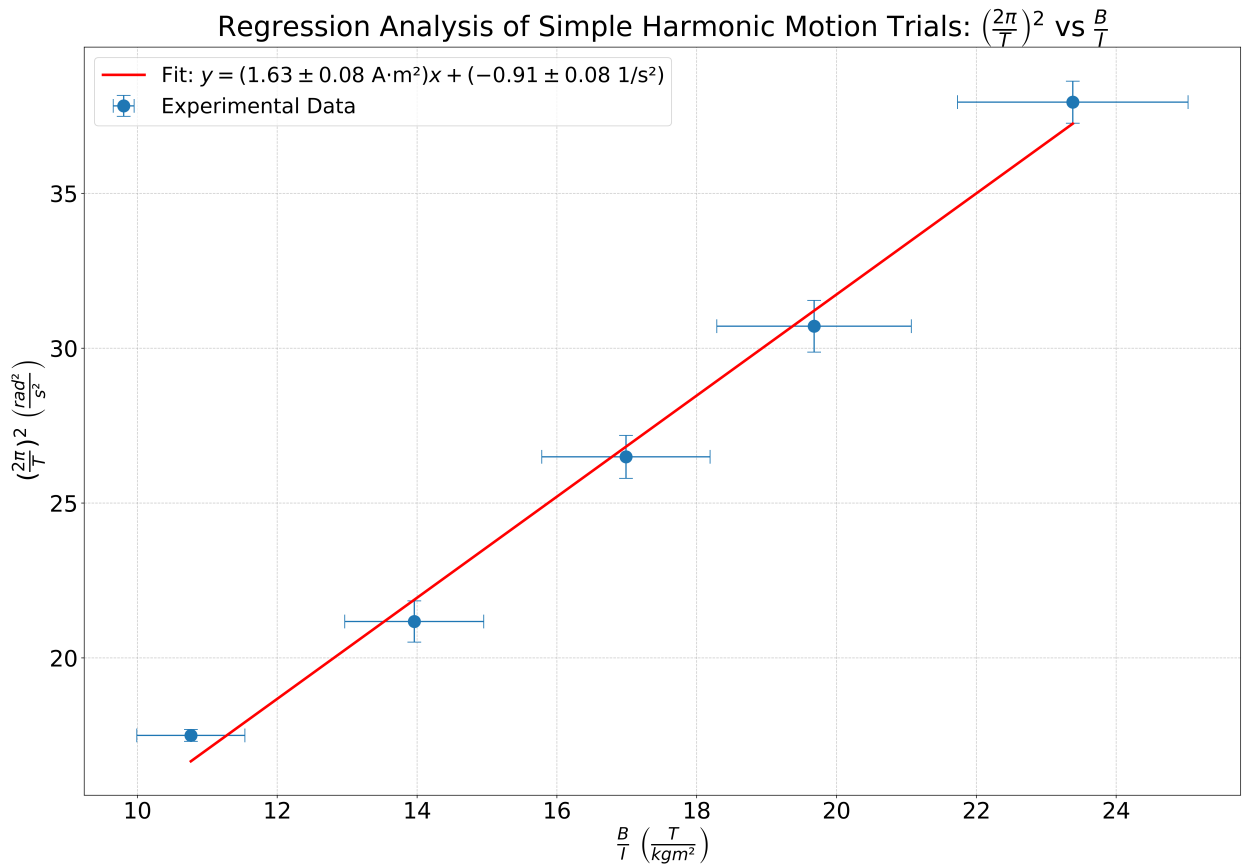

Figure 3: A plot of $\left(\frac{2\pi}{T}\right)^2$ against $\frac{B}{I}$ for the simple harmonic motion experiment. The linear regression fit is shown in red, with associated data points from Table 1 in blue. Origin not shown.

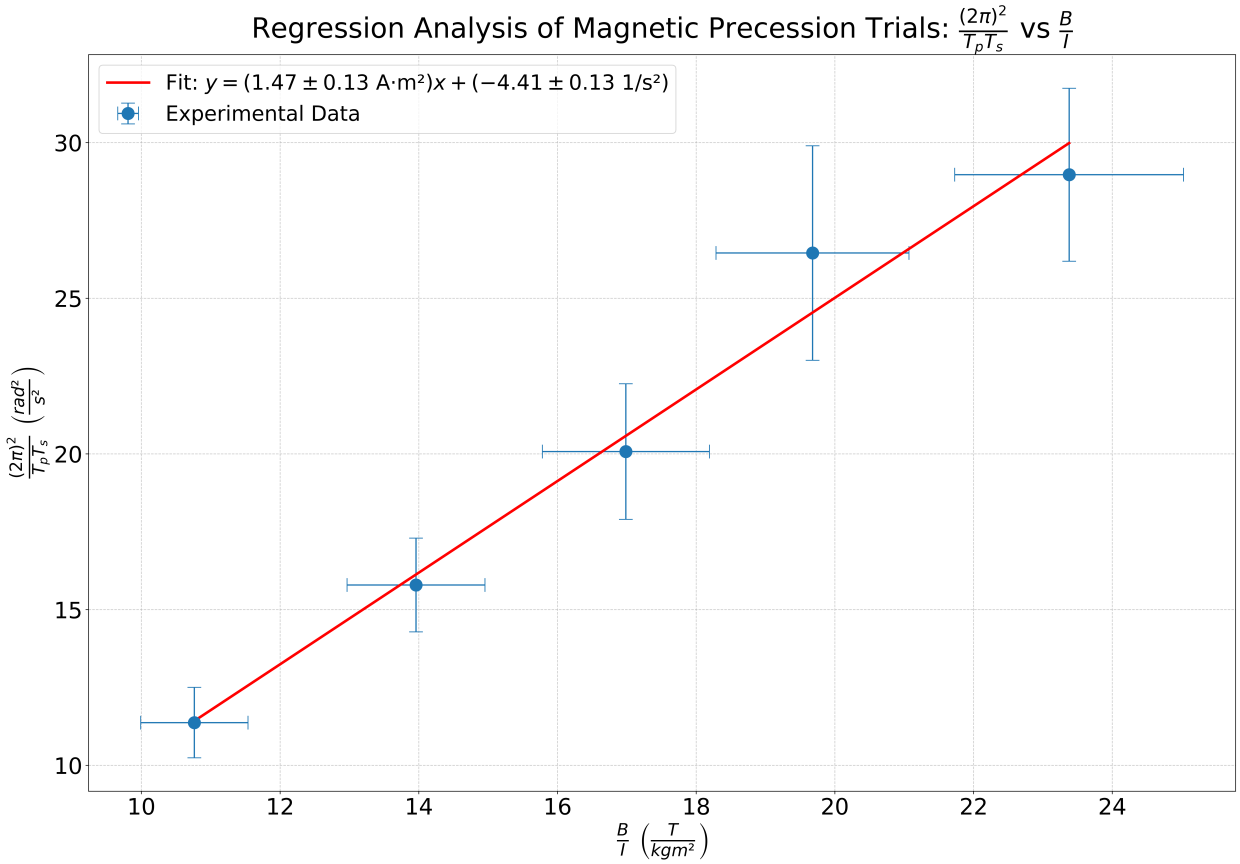

Figure 4: A plot of $\frac{(2\pi)^2}{T_p T_s}$ against $\frac{B}{I}$ for the magnetic precession experiment. The linear regression fit is shown in red, with associated data points from Table 1 in blue. Origin not shown.

A pointwise analysis of the data is also performed, yielding one value of μ per trial per experiment (harmonic motion and precession). These values are organized in Table 2. A mean and standard-deviation-of-mean analysis is performed, yielding μ-values of (1.577 ± 0.021) A $\cdot$ m^2 and (1.19 ± 0.05) A $\cdot$ m^2 from the harmonic motion and precession experiments, respectively. These are not consistent with each other. However, the value of μ from the pointwise analysis for the harmonic motion experiment is consistent with the values of μ from the linear regression analysis–suggesting that the precession experiment deserves futher attention. Pointwise values of μ in comparison with the mean values are organized in Figures 5 and 6.

Current (A) ($\pm$ 0.01 A)	μ, SHO (A $\cdot$ m^2)	μ, Precession (A $\cdot$ m^2)
2.00	1.62 ± 0.12	1.24 ± 0.15
1.75	1.56 ± 0.12	1.34 ± 0.20
1.50	1.56 ± 0.12	1.18 ± 0.15
1.25	1.52 ± 0.12	1.13 ± 0.13
1.00	1.63 ± 0.12	1.06 ± 0.13

Table 2: Values of the dipole strength μ computed per current value per experiment.

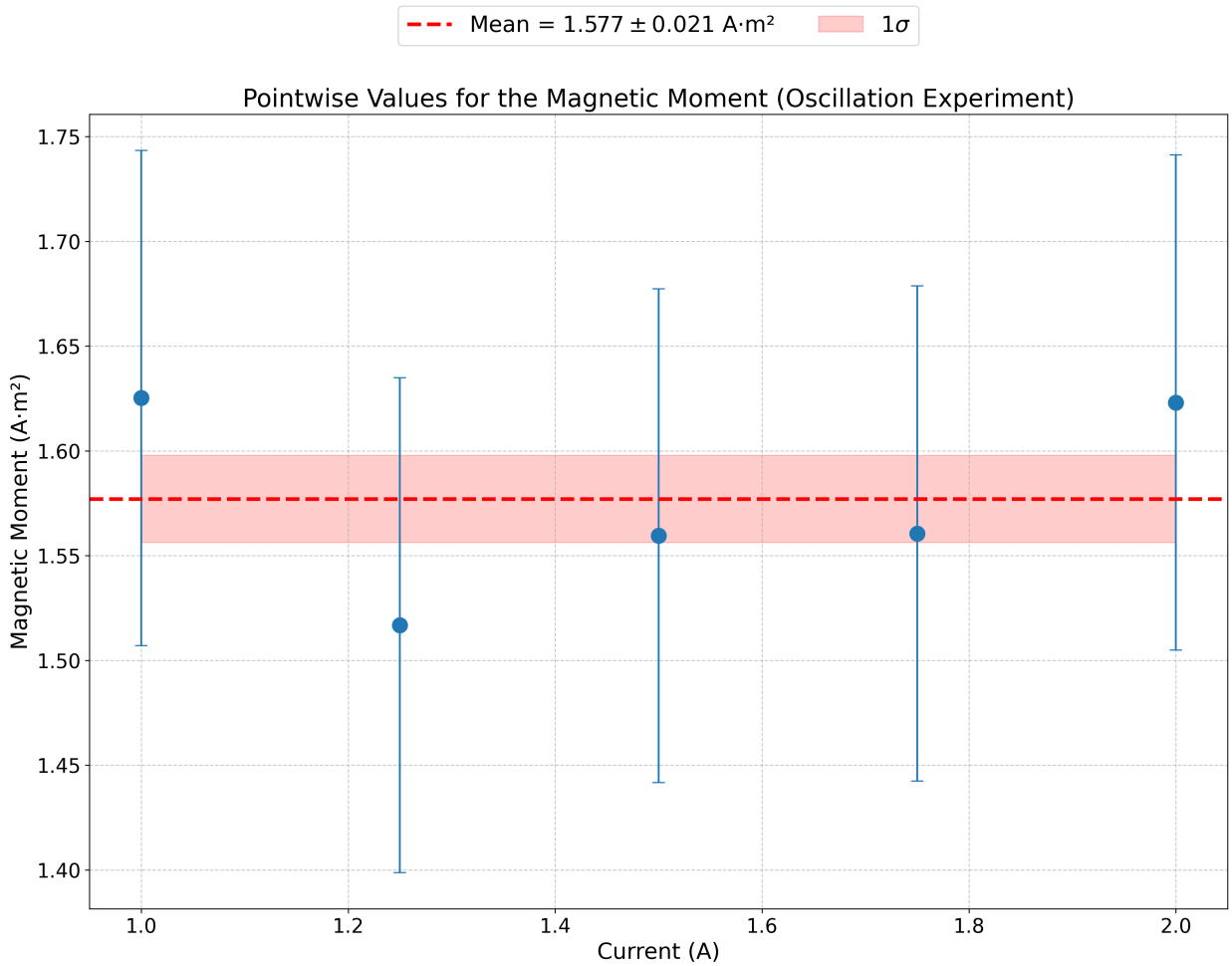

Figure 5: A plot resulting from the pointwise analysis of the simple harmonic motion experiment. The best estimate (mean) is indicated with a red dashed line, with the 1σ SDOM uncertainty interval shaded in red. Points from Table 2 are shown in blue. Origin not shown.

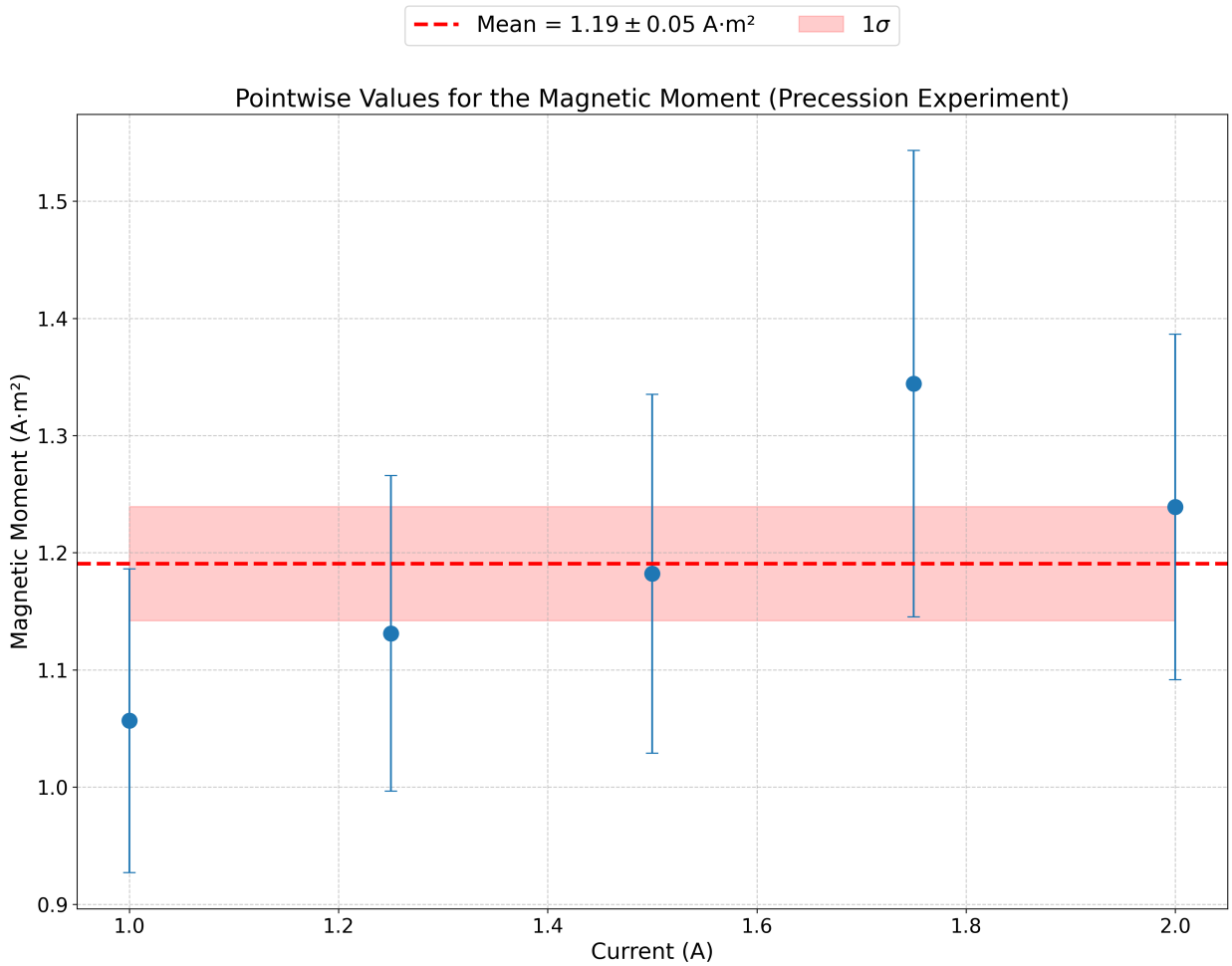

Figure 6: A plot resulting from the pointwise analysis of the magnetic precession experiment. The best estimate (mean) is indicated with a red dashed line, with the 1σ SDOM uncertainty interval shaded in red. Points from Table 2 are shown in blue. Origin not shown.

4 Conclusion and Discussion

Three experiments are conducted to investigate the behaviors of magnetic dipoles under the influence of torques produced by external magnetic fields. The first two obtain experimental values of the dipole strength μ from the simple harmonic motion and magnetic precession of the dipole in the magnetic field. The simple harmonic motion experiment yields $\mu = (1.63 \pm 0.08)\ \mathrm{A} \cdot \mathrm{m}^2$ from linear regression and $\mu = (1.577 \pm 0.021)\ \mathrm{A} \cdot \mathrm{m}^2$ from pointwise analysis. The magnetic precession experiment yields $\mu = (1.47 \pm 0.13)\ \mathrm{A} \cdot \mathrm{m}^2$ from linear regression and $(1.19 \pm 0.05)\ \mathrm{A} \cdot \mathrm{m}^2$ from pointwise analysis.

The inconsistency between the magnetic precession pointwise results and the three other results encourages investigation into sources of inaccuracy in the precession experiments. The method used for obtaining ω_s readings from the tachometer is suspect; it is possible that the decay of the dipole's speed is not strictly linear, causing a deviation of an ideal estimate of ω_s from the average value of ω_s on the time interval of measurement. The dipole itself also disturbs the magnetic field produced

by the coils once introduced, potentially invalidating readings of the magnetic field strength made before placing the dipole in the platform. Measurements of the precession period also could have been made more robust by averaging over several trials or by using a more precise method for timing, such as a video camera instead of a clock.

Across all experiments, the gravitational torque acting on the dipole contributes to discrepancies between the experimental and true values of μ. Due to the cavity created by embedding the magnet in the dipole, the dipole has a non-uniform mass distribution and thus the gravitational force does not act directly along the dipole's axis of rotation or oscillation. Earth's magnetic field also contributes a slight horizontal magnetic field component at the dipole's position, as do imperfections in the alignment of the coils with the horizontal plane.

A third experiment investigates the spin flip mechanism of the dipole. Qualitatively, the torque produced by the transverse magnetic field is observed to be in line with the theoretical prediction, and the dipole's orientation is successfully flipped.

References

[1] Randy Harris. *Modern Physics*. Pearson, Harlow, 2014.

[2] William F. Trench. *Elementary Differential Equations*. Trinity University, San Antonio, TX, 2013.

[3] Hugh D Young and Roger A Freedman. *University Physics: with Modern Physics*. Harlow Pearson Education Limited, 15 edition, 2020.